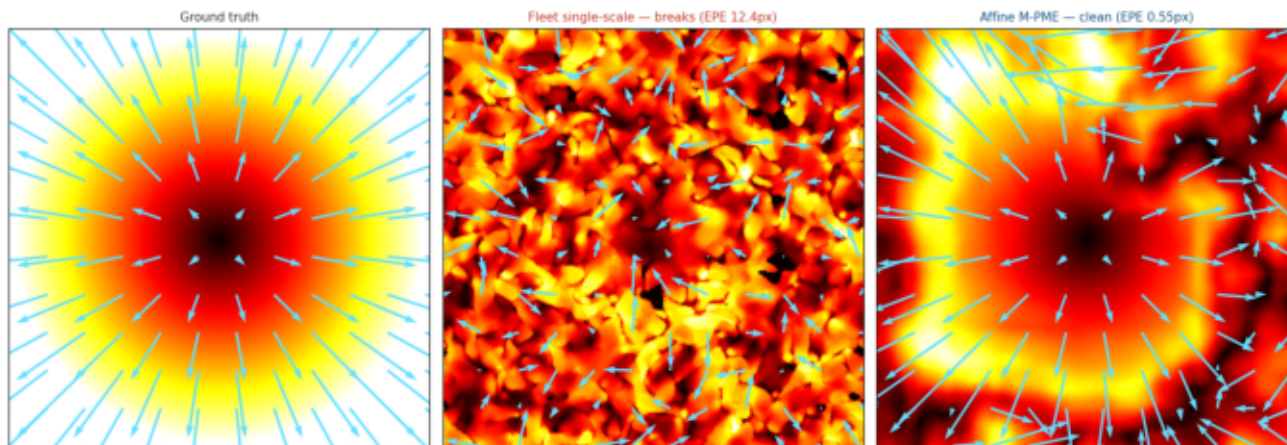


The Affine Ceiling

What deterministic motion estimation solves — and what it doesn't.



A 22% scaling. Left: ground-truth radial flow. Middle: single-scale phase flow collapses. Right: affine multi-scale (M-PME) recovers it to 0.55 px median error.

Phase-based motion estimation is deterministic and certifiable — the kind of algorithm you can ship into a safety-critical system. The open question is not whether it is trustworthy, but how far it reaches. We measured it against pixel-exact ground truth, across three implementations of increasing fidelity.

The finding overturns the intuitive story. A single-scale estimate is capped at half a wavelength and fails on large motion. But a multi-scale, affine, coarse-to-fine method — the 2026 state of the art — solves large scaling and rotation to SUB-PIXEL accuracy. The real ceiling is not motion size. It is the affine assumption itself.

Why determinism is the real constraint

Vehicles, surgical robots and aircraft live under functional-safety standards — ISO 26262, DO-178C, IEC 62304 — that demand a provable worst case. A learned depth or optical-flow network offers neither a bound nor a repeatable output: the same scene in different light returns different numbers. It is probabilistic by construction — acceptable for a photo app, disqualifying for a braking decision.

Classical signal processing has the opposite profile: every output is reproducible and analyzable. The only question that matters is how accurate it is, and over what range of motion.

02 – THE SINGLE-SCALE LIMIT

A limit you can derive on paper

Phase-based estimation reads displacement from the phase shift of a band-pass filter. The Fourier shift theorem makes that shift linear in the displacement:

$$\Delta\varphi = 2\pi k \delta \quad \longrightarrow \quad \delta_{\max} = \lambda/2$$

phase is unique only in $(-\pi, \pi]$;
past half a wavelength it wraps

Beyond half a filter wavelength the phase aliases and the answer is silently wrong. Fleet & Jepson characterized this in the early 1990s, down to which measurements to trust and which to reject [1][2]. A SINGLE-scale method therefore has a hard ceiling — and it is small. That ceiling is real. The question is whether it is fundamental. It is not.

03 – THREE IMPLEMENTATIONS

From a hard ceiling to none

- Fleet (single-scale). One octave band, measured local frequency, singularity masking. Bounded to $\delta < \lambda/2$ — our reference for the limit.
- Reimpl (no paper). A multi-scale Gaussian pyramid with per-pixel translation and rewrapping, built before reading the source paper. Extends the range, but leaves boundary and periphery outliers.
- M-PME (faithful). The 2026 method of Li et al. [5]: a Gaussian pyramid with an AFFINE 6-parameter motion model fit over a window (Farnebaeck-style), confidence weighting, four Gabor directions, coarse-to-fine superposition. Scale and rotation ARE affine — so this models them exactly.

Endpoint error against pixel-exact ground truth

We warp a broadband texture by an exactly known scaling, giving a pixel-exact ground-truth flow, then sweep the scale and measure median endpoint error (EPE).

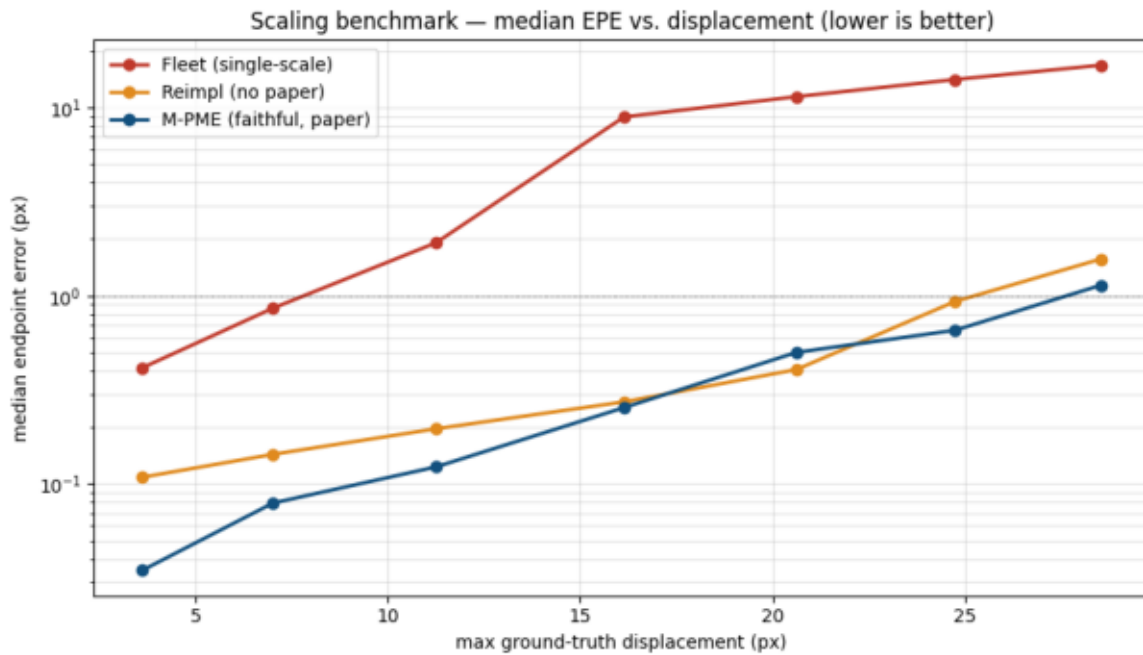


Fig. 1 Median EPE vs. displacement (log scale). Single-scale fails past ~11 px; both multi-scale methods stay sub-pixel; the affine M-PME is cleanest.

max displacement	Fleet	Reimpl	M-PME
7 px	0.86	0.14	0.08
16 px	8.93	0.27	0.25
25 px	14.11	0.93	0.66

median EPE in px — lower is better

The recovered fields, side by side

Ground truth against all three methods, at increasing scale (top to bottom). Single-scale disintegrates; the per-pixel reimplementation is close in the median but throws visible outlier vectors; the affine M-PME tracks the radial field cleanly.

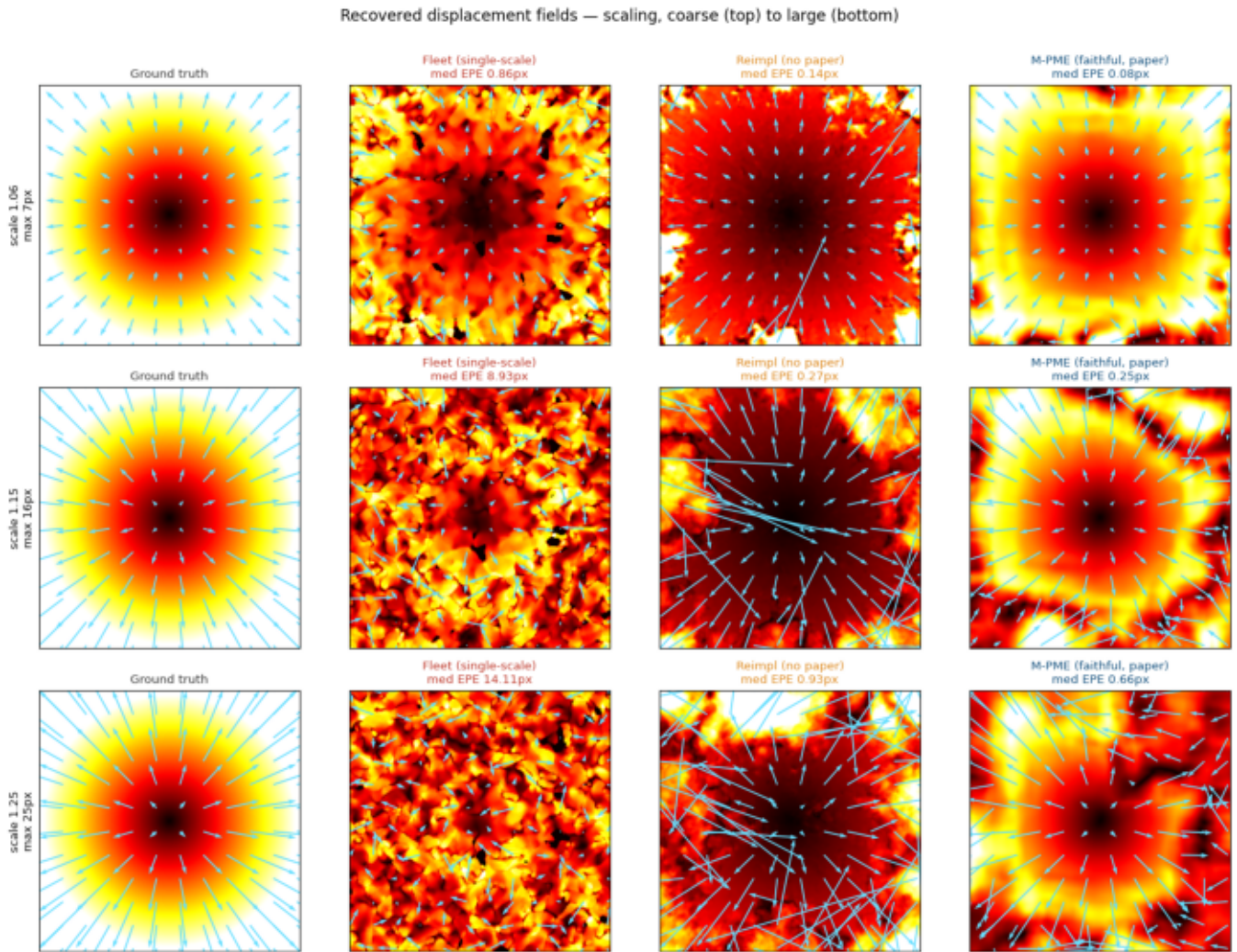


Fig. 2 Displacement fields. Columns: ground truth, Fleet single-scale, per-pixel reimplementation, affine M-PME. Rows: scale 1.06 / 1.15 / 1.25.

Affine is solved. Real 3D is not.

Deterministic phase flow solves large affine motion — translation, rotation, scale — to sub-pixel accuracy. The ceiling is the affine assumption, not the motion size.

Credit where due: the multi-scale, affine, coarse-to-fine approach genuinely removes the small-motion ceiling. Li et al. track a rotating turbine blade through hundreds of pixels of motion at sub-pixel error, and our faithful reimplementation reproduces sub-pixel accuracy on large scaling.

But M-PME assumes the motion is AFFINE within each window. Their only tests are a single rigid object on a black background — the easiest possible affine case. Dense 2D->3D reconstruction of a real scene is not affine: it has per-pixel depth parallax, occlusion boundaries where motion is discontinuous, and independently moving objects that share a window. That is where an affine-per-window model must break — and it is exactly what depth reconstruction depends on.

So the frontier moves. Not 'deterministic can only do small motion' — that is false. The open problem is deterministic, certifiable, PER-PIXEL, NON-AFFINE motion for real 3D scenes. That is what ImplicitKalman is for.

METHOD NOTE

Single controlled family: synthetic broadband texture warped by an exact scaling (pixel-exact GT). EPE reported as the median over a central crop (robust to boundary-window artefacts), matching the high-confidence evaluation the paper uses. M-PME is our faithful reimplementation of the described method. All figures reproducible from source.

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